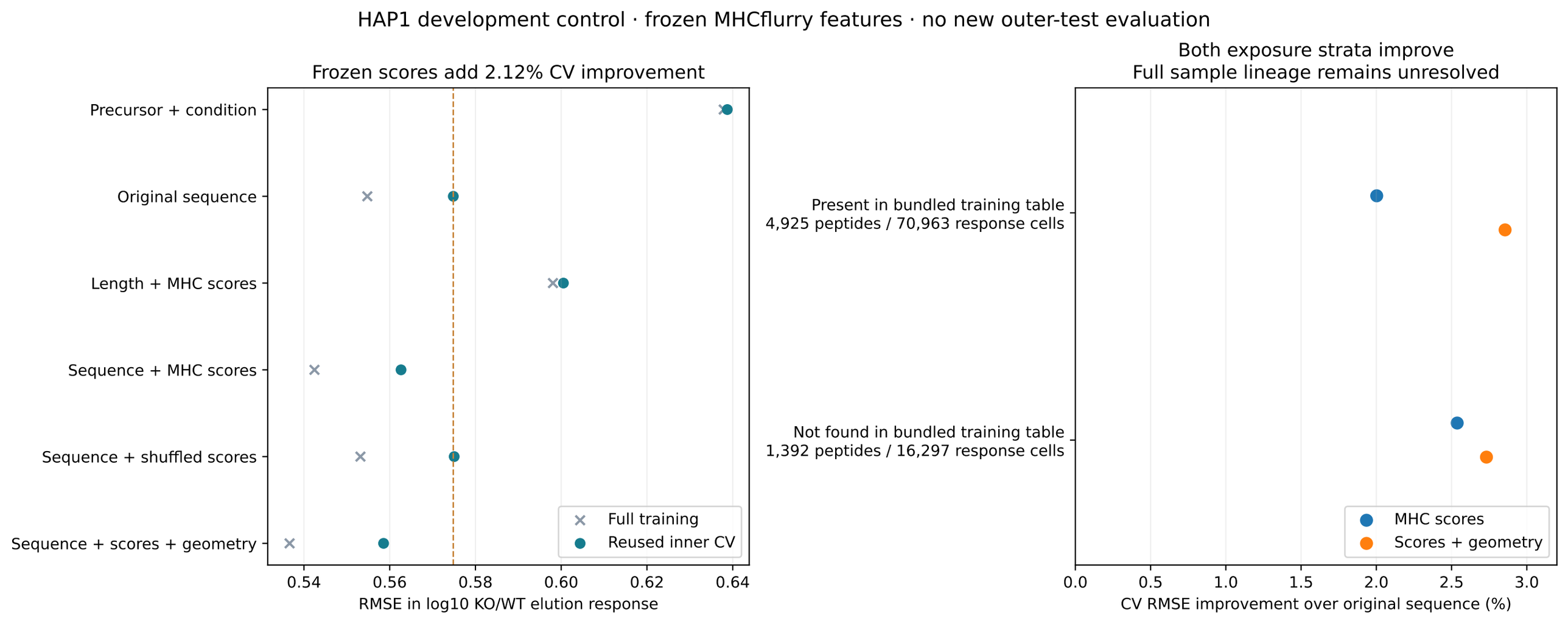
HAP1 development control · frozen MHCflurry features · no new outer-test evaluation
Frozen scores add 2.12% CV improvement
Precursor + condition
Original sequence
Length + MHC scores
Sequence + MHC scores
Sequence + shuffled scores
Sequence + scores + geometry
Full training
Reused inner CV
0.54
0.56
0.58
0.60
0.62
0.64
RMSE in log10 KO/WT elution response
Both exposure strata improve
Full sample lineage remains unresolved
Present in bundled training table
4,925 peptides / 70,963 response cells
Not found in bundled training table
1,392 peptides / 16,297 response cells
MHC scores
Scores + geometry
0.0
0.5
1.0
1.5
2.0
2.5
3.0
CV RMSE improvement over original sequence (%)